LIABILITY-DRIVEN & GOAL-BASED INVESTING

EDHEC Business School · M2

Group Assignment — Case Study

Engineering a Customised Investment Solution for Retirement

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INTRODUCTION — OBJECTIVES OF THE INVESTMENT SOLUTION

The goal is to help the client prepare for retirement, which will start in 10 years on Nov 1, 2024. More precisely, we want to generate a solution providing $50,000 annual replacement income for 20 years after the retirement date, i.e. with a last payment on Nov 1, 2043. We will respect the budget constraint that corresponds to a 50% Funding Ratio with respect to the target, meaning that the client is able to fund half of the 20 annual replacement income cash flows with no expected additional contribution. The risk budget of the client corresponds to 80% of their current funding status, i.e. $20,000 annually. An expected inflation of 2% per annum needs to be taken into account.

The key metrics to consider are therefore:

Aspirational Goal (Targeted Replacement Income): $50,000 p.a.

Essential Goal (Minimum Required — Risk Budget): $20,000 p.a.

Initial Funding Ratio: 50%, i.e. $25,000 p.a.

Expected Inflation: 2% p.a.

Accumulation Period: 10 years, from Oct 31, 2014 to Nov 1, 2024

Decumulation Period: 20 years, from Nov 1, 2024 to Nov 1, 2043

The goal is to design an investment strategy aimed at maximizing the probability of reaching the Aspirational Goal, while ensuring a 100% probability of meeting the Essential Goal. Therefore, a safe component, the Goal-Hedging Portfolio, is needed to hedge the annual $20,000. Another component aimed at offering upside potential, the Performance-Seeking Portfolio, will be considered to offer diversification and target the annual $50,000. Finally, an optimal dynamic allocation (offering insurance) between these two components will be implemented to ensure an optimized investment strategy.

I THE GOAL-HEDGING PORTFOLIO

1.1 The Retirement Bond — Definition

The chosen Goal-Hedging Portfolio is the Retirement Bond. It is a financial security aimed at answering retirement income goals by offering a constant annual cash flow for each year considered after retirement. In this case, the Retirement Bond will therefore offer 20 cash flows for each of the 20 years of retirement.

As it remains fictitious as of today, it needs to be replicated from existing bonds and the sovereign yield curve. The price of the Retirement Bond follows the following formula:

$$RB_t = \Sigma_{i=0}^{19} (1 + Inf)^{(T+i)} \cdot exp[-((T+i) - t) \cdot r_t(T+i)-t]$$

with 1 being the $1 normalised value of the Retirement Bond cash flows, Inf being the expected inflation (2% in this case), T the retirement date (Nov 1, 2024), and $r_t,(T+i)-t$ the zero-coupon discount rates. To correctly scale the investment, the RB cash flow needs to be adjusted according to the essential goal we are looking to hedge. The previous formula is therefore adjusted, with C = $20,000:

$$RB_t = C \cdot \Sigma_{i=0}{}^{19} (1 + Inf)^{(T+i)} \cdot exp[-((T+i) - t) \cdot r_t,(T+i)-t]$$

1.2 Interest Rate Simulations

In order to replicate the Retirement Bond, we first need to simulate scenarios for the short rate according to the Vasicek model, through 1,000 Monte Carlo simulations, and using a reflection principle to erase unrealistic scenarios with negative rates. Here is a graph presenting the result of our simulations :

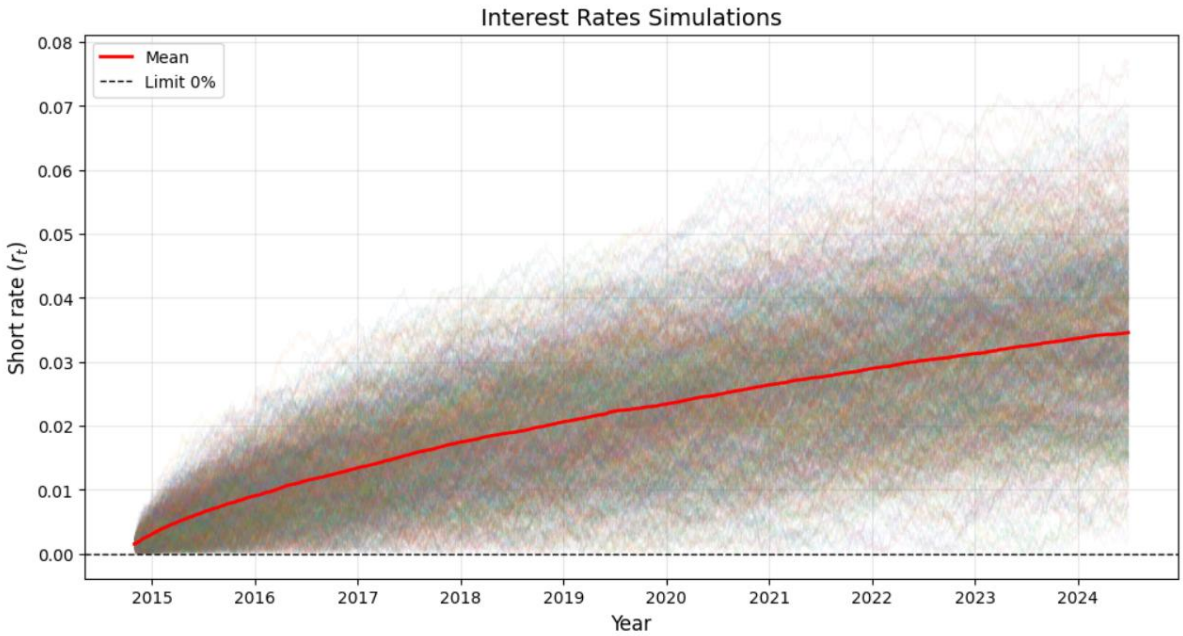


Figure 1.1 — Vasicek Short Rate Monte Carlo Simulations (1,000 scenarios). The simulations are consistent with the available Zero Coupon Yields from Oct 2014 to Nov 2024, with a long-term mean estimated around 5%.

The simulations are consistent with the available Zero Coupon Yields from the 10/31/2014 to the 11/01/2024, with a long term mean estimated around 5% according to the model used.

1.3 Goal-Hedging Portfolio Replication

A simple approach is a static cash-flow matching approach. Using the simulated Zero-Coupon Yield Curve data, we can match the desired cash flows by considering the RB as a set of Pure Discount Bonds with maturities ranging from T = 10 (retirement date) to T = 30 (last payment date). At each payment date, the RB will pay a $20,000 inflation-adjusted cash flow, and the price RB_t will represent the Present

Value of these cash-flows after discounting.

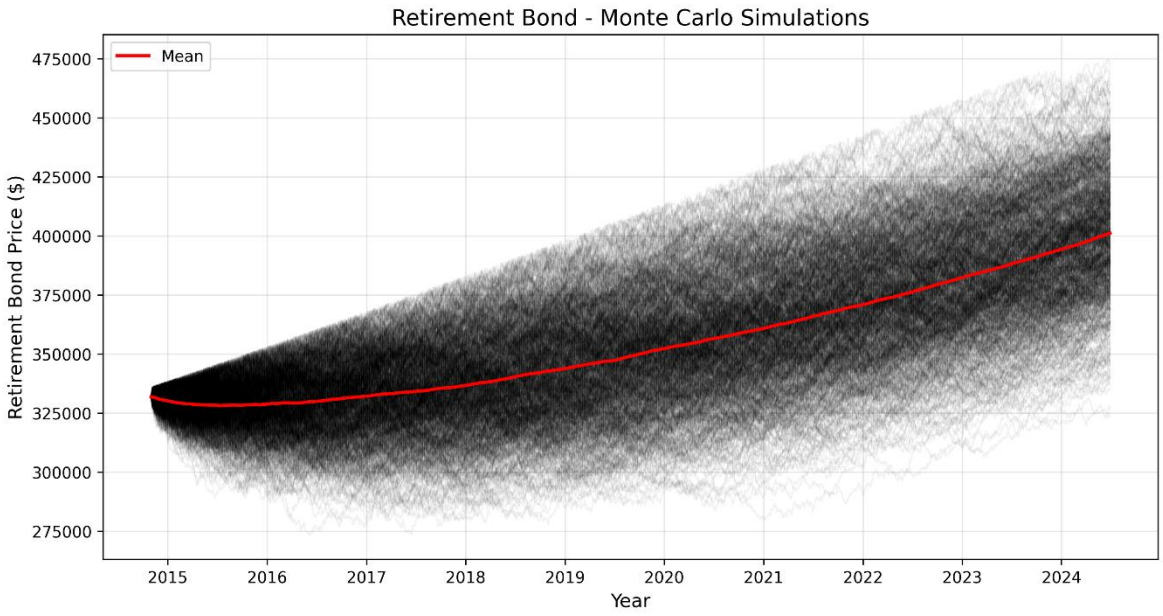


Figure 1.2 — Retirement Bond Price Paths across 1,000 Monte Carlo Scenarios.

Our simulations give a Retirement Bond priced at $331,922.05 today. This result is consistent with the funding of the client. In fact, as of 2014 in dollar terms:

Objective	Annual Replacement Income	Present Value (Oct 2014)
Current Funding	$25,000 per retirement year	**$414,902.56**
Essential Goal	$20,000 per retirement year	**$331,922.05**
Aspirational Goal	$50,000 per retirement year	**$829,805.12**

Table 1.1 — Summary of Goal Values at inception (Oct 2014).

The value of the Retirement Bond corresponds to the essential goal as it is designed to secure its set of cash flows and ensure that the minimum requirement of the client is met.

1.4 The Sub-Optimal Benchmark — A Constant Maturity Bond Index

In order to provide better insights into the welfare gains provided by the Retirement Bond, we will compare its performance against a sub-optimal benchmark, a Constant Maturity Bond (CMB) Index.

A CMB index re-weights a portfolio to keep a fixed maturity exposure through rolling liquid Treasury Bond instruments. It offers a simple representation of a "safe" instrument with continuous rebalancing. It is a useful benchmark, as opposed to our chosen GHP (Retirement Bond), as it is a readily investable proxy for the safe component. Even so, it remains sub-optimal to answer the client's retirement goal for three reasons.

First, there is a liability mismatch: the CMB targets a constant duration exposure and does not explicitly match cash-flows, while the RB is explicitly constructed to replicate the timing and inflation-adjusted amounts of the retirement goal cash-flows. Its PV directly targets the PV of the given liabilities.

Second, it imposes reinvestment risk as it requires continuous rolling into a target maturity, exposing the investor to future yield curve movements. The RB pins down payments and minimizes reinvestment risk.

Third, the CMB market value responds differently to shifts and twists of the yield curve, producing a different Funding Ratio volatility compared to an RB explicitly matched to liability maturities.

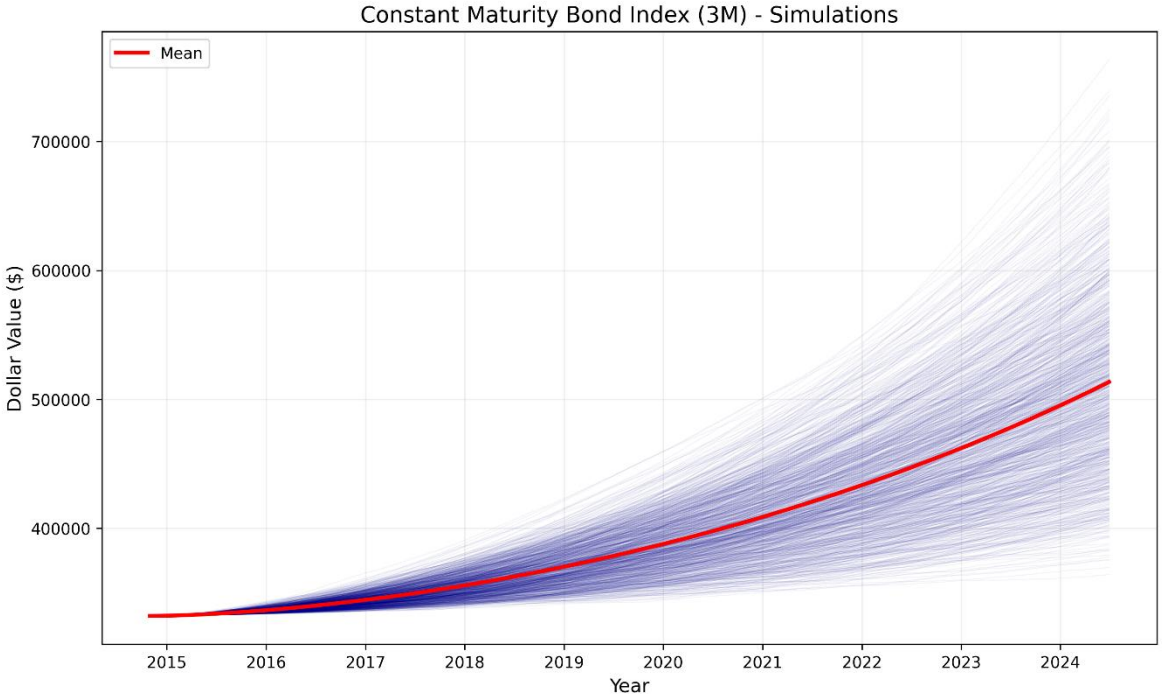


Figure 1.3 — Constant Maturity Bond Index (3-Month) Simulations (1,000 scenarios). Terminal mean value: $513,503.21 after 10 years.

1.5 Performance Comparison

A variable of interest to assess the performance of the strategy with respect to the client's retirement goals is the Funding Ratio, defined as:

$$FR_t = A_t / L_t$$

where A_t is the total market value of the client's portfolio at time t (their assets) and L_t the total value of their liabilities. Put differently, L_t represents the client's retirement goal, the cash-flows they require for the decumulation period. The FR is essential to manage risk exposures accordingly to the given risk budget.

Thanks to our previous simulations, we can compare the terminal values of this funding ratios and their volatility, the final distribution in 2024 of the assets (the Retirement Bond or the benchmark), and more importantly the probability of reaching the essential goal, i.e :

$$\hat{P}_{essential} = (1/N_{sim}) \Sigma_{i=1}^{N} \mathbf{1}\{ FR_{t,i} \geq 1 \}$$

Our results demonstrate a greater volatility, and therefore greater risk for the essential goal, in the case of the Constant Maturity Bond Index. The Retirement Bond is designed to perfectly match the expected liabilities, and therefore represents a real safe component in the strategy as opposed to the inefficient benchmark. The final distributions at T=10 are shown below:

	Retirement Bond ($)	CMB 3M ($)
Mean	401144.392920	513503.206010
Std Dev	25258.820755	70457.736193
Min	324125.327999	363958.404905
Max	472135.583229	763701.333297
5th Percentile (VaR 95%)	360781.014665	411523.533218
95th Percentile	442353.844439	640586.391822
Skewness	-0.024794	0.526690
Kurtosis	-0.198360	-0.006921

--- RETURN METRICS (LOG-SCALE) ---

	RB Log-Returns	CMB 3M Log-Returns
Mean	0.187426	0.427159
Std Dev	0.063310	0.135060
Min	-0.023770	0.092139
Max	0.352366	0.833277
5th Percentile (VaR 95%)	0.083371	0.214966
95th Percentile	0.287210	0.657484
Skewness	-0.197561	0.199774
Kurtosis	-0.116188	-0.390116

Figure 1.4 — Terminal Distribution Statistics: Retirement Bond (top) and CMB 3M (bottom).

The Retirement Bond provides on average $401,144.39 in 2024, which corresponds to an average return of 18.74% in 10 years, i.e. around 1.70% per year. The standard deviation is low, $25,258.82 (around 6.33%) and corresponds to the adaptation of the Retirement Bond to the liabilities. The distribution is slightly negatively skewed, but the skewness remains close to 0. The distribution is close to be Gaussian, as confirmed by the low excess kurtosis. These higher moments prove the efficiency of the Retirement Bond, with a low risk and the absence of extreme tail events. We also note the narrower gap of $148,010.25 between the maximum scenario value and the minimum one, as opposed to $399,742.93 in the case of the Constant Maturity Bond Index.

Although the Constant Maturity Bond Index (3 months) exhibits a higher average nominal return (42.71% vs 18.74%), it fails as a hedging instrument for the retirement goal. Its wealth distribution at T=10 is nearly three times as volatile as the Retirement Bond's ($70,457 vs $25,258). Moreover, the dispersion of its return is more than twice greater than the returns of the Retirement Bond (13.50% vs 6.33%). The skewness is significantly higher (0.52) as well as the gap between the worst and the best case scenarios. Therefore, the Constant Maturity Bond Index provides no certainty on future cash flows, making it a poor "safe" instrument for the client's goal.

The following table focuses on the Funding Ratio under the Constant Maturity Bond Index :

	CMB Funding Ratio
Mean	1.293490
Std Dev	0.247133
Min	0.802619
Max	2.243232
5th Percentile (VaR 95%)	0.948886
95th Percentile	1.737620
Skewness	0.675837
Kurtosis	0.372055

Probability of reaching the Essential Goal (CMB) : 90.00%
Funding Ratio annualized volatility (CMB) : 3.99%

Table 1.2 — Funding Ratio Statistics: CMB 3M Benchmark at T=10.

Despite a high mean (1.29%), the Funding Ratio exhibits some volatility (3.99%) and a 5% worst case scenario under the minimum level of 1 (100%) to secure the essential goal. In fact, as opposed to the Retirement Bond, the benchmark provides a probability of securing the essential goal of 90%.

On the other hand, our computations show that the Retirement Bond guarantees a probability of 100% across all scenarios, as well as an annualized Funding Ratio volatility of 0%, making it a better and optimal hedging component for our investment solution. To better illustrate this idea, the following graph presents the returns of the Funding Ratios across all the scenarios :

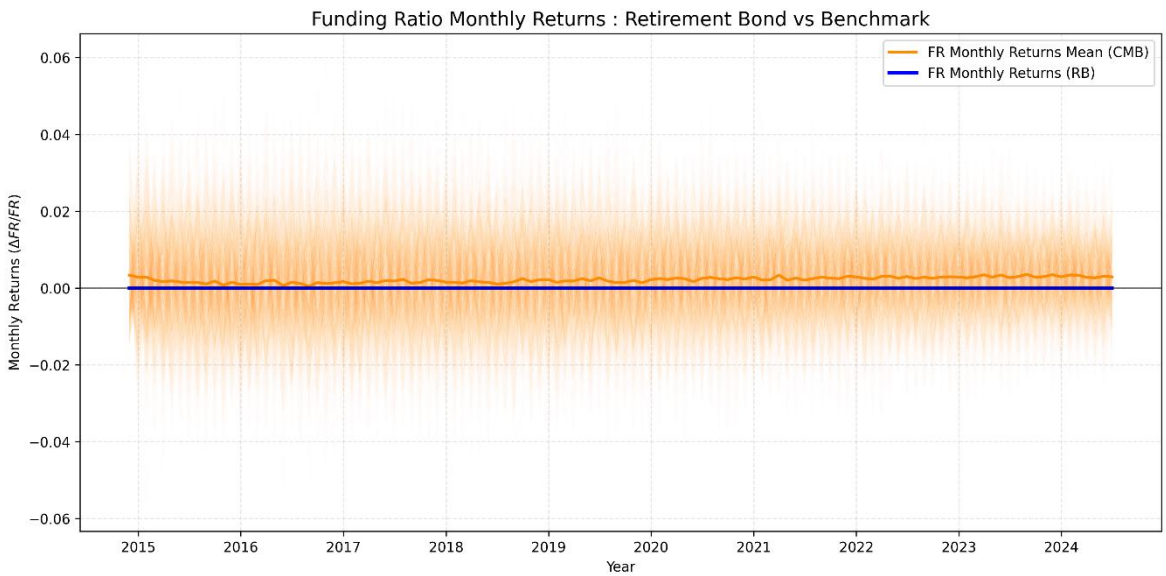


Figure 1.5 — Monthly Funding Ratio Returns across all scenarios: Retirement Bond (flat at 0, zero variability) vs CMB 3M (wider oscillations). The flat FR return for the RB confirms perfect liability matching at all times.

This graph presents the returns of the monthly Funding Ratios across scenarios for the benchmark and the Retirement Bond. The Funding Ratio for the Retirement Bond is flat at 0, which is another proof of the efficiency of this instruments. The variability of the Funding Ratio is 0, meaning that we constantly matches the required liabilities for the essential goal. Nevertheless, the Constant Maturity Bond Index

exhibits a greater volatility around 0, demonstrating the lack of certainty of the Funding Ratio across time and scenarios.

II THE PERFORMANCE-SEEKING PORTFOLIO & DYNAMIC ALLOCATION

From a welfare perspective, the objective is not to maximise expected portfolio returns, but rather to increase the probability of achieving the aspirational retirement income while ensuring that the essential consumption floor remains protected across all market scenarios. While the Goal-Hedging Portfolio ensures the protection of the essential retirement income, it does not allow the investor to reach the aspirational goal of $50,000 per year. We therefore introduce a second building block: the Performance-Seeking Portfolio (PSP), combined with a dynamic CPPI allocation mechanism.

The following diagram summarises the complete investment architecture:

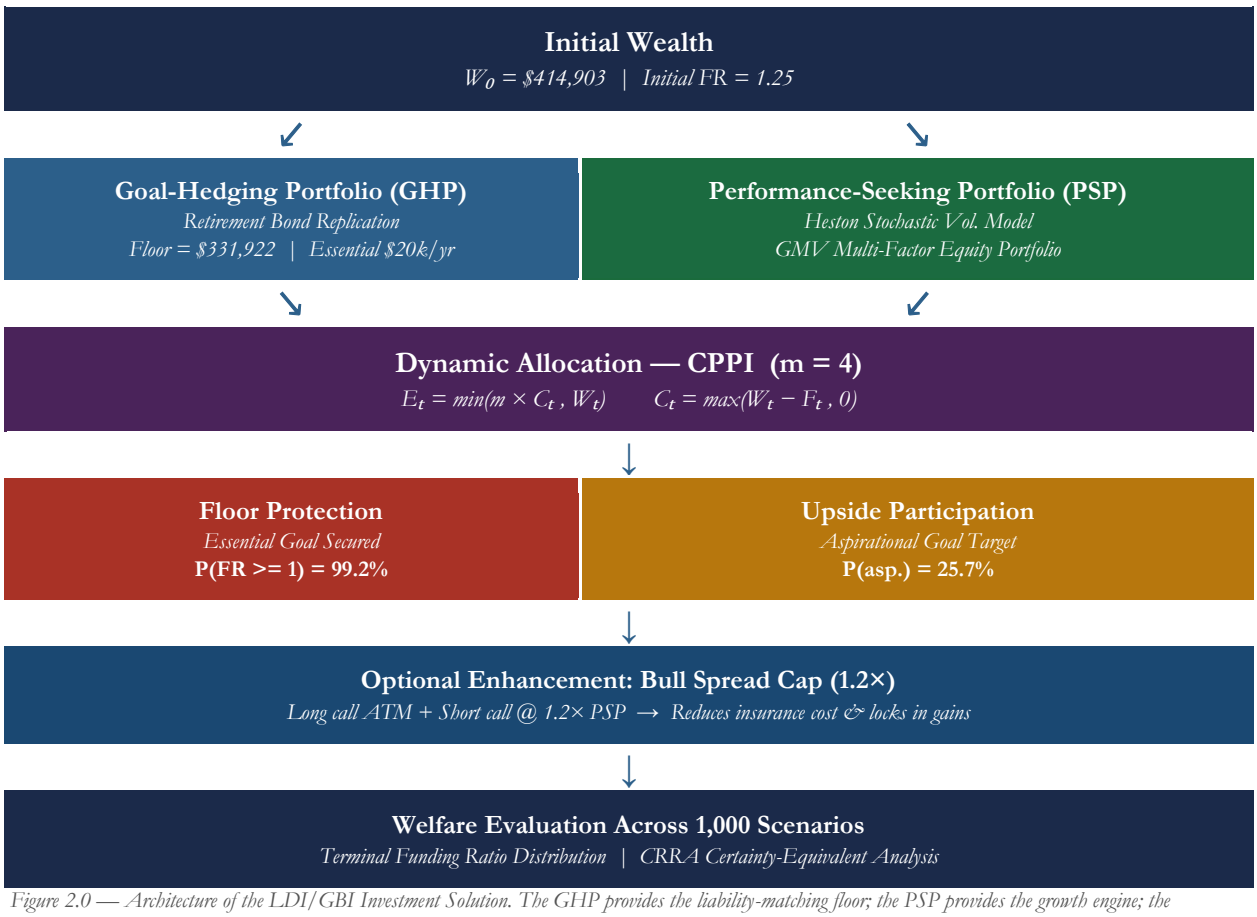


Figure 2.0 — Architecture of the LDI/GBI Investment Solution. The GHP provides the liability-matching floor; the PSP provides the growth engine; the CPPI dynamically allocates between the two in proportion to the available risk budget (cushion).

The strategy is structured around two complementary building blocks. The Goal-Hedging Portfolio (GHP), detailed in Part I of this report, is the Retirement Bond, a set of zero-coupon bonds that precisely replicates the cash flows of the essential retirement income. It constitutes the liability-matching floor that the CPPI must preserve at all times. The Performance-Seeking Portfolio (PSP), developed in Sections II and III below, is a multi-factor equity portfolio simulated under the Heston stochastic

volatility model. It provides the growth engine that drives wealth above the floor toward the aspirational goal. The CPPI dynamically allocates between the two in proportion to the available cushion, ensuring the floor is never breached while maximising upside participation. Finally, the bull spread overlay reduces the cost of insurance by capping redundant upside at 1.2x the PSP level and recycling that premium into the safe sleeve.

2.1. The Equity Model — Heston Stochastic Volatility

In order to model the risky building block (the Performance-Seeking Portfolio, PSP), we adopt the Heston (1993) stochastic volatility model. While a simple Geometric Brownian Motion (GBM) with constant volatility offers analytical tractability, it fails to capture key empirical features of equity returns, namely volatility clustering, fat tails, and the leverage effect (i.e., the negative correlation between equity returns and volatility changes). The Heston model addresses these shortcomings by allowing the variance process to be stochastic and mean-reverting, while preserving a degree of analytical tractability for pricing and calibration.

The model is governed by a two-factor diffusion system under the real-world measure P:

$$dS_t = \mu S_t dt + \sqrt{V_t} S_t \, dW_t^S$$
$$dV_t = \kappa(\theta - V_t)\, dt + \sigma_v \sqrt{V_t}\, dW_t^V$$
$$d(W^S, W^V)_t = \rho \, dt$$

where S_t is the equity price, V_t is the instantaneous variance, μ is the annualised expected return, κ is the mean-reversion speed of the variance, θ is the long-run variance, σ_v is the volatility-of-volatility (vol-of-vol), and ρ is the correlation between the two Brownian motions.

The parameters of the model were calibrated as follows. The long-run drift μ = 8% per annum reflects the historical equity risk premium. The long-run variance θ = 0.04 corresponds to a long-run annualised volatility of 20%, consistent with historical S&P 500 data. The mean-reversion speed κ = 2.0 was estimated via MLE on the VIX time series, ensuring that short-term variance spikes are absorbed within a plausible time frame. The vol-of-vol σ_v = 0.20 was fit to the shape of the implied volatility surface, and the leverage correlation $\rho = -0.70$ was estimated from the historical co-movement between S&P 500 returns and the VIX, a standard empirical finding. The initial variance V_0 = 0.04 is set equal to its long-run mean, starting the simulation at a stationary point.

An important regularity condition for the Heston model is the **Feller condition**: $2\kappa\theta > \sigma_v^2$. With our parameters, $2 \times 2.0 \times 0.04 = 0.16 > 0.04 = \sigma_v^2$, so this condition is satisfied, guaranteeing that the variance process V_t remains strictly positive almost surely. We further enforce a full truncation scheme in the numerical discretisation to prevent any negative variance draws arising from Euler-Maruyama discretisation error.

Monte Carlo simulations were conducted over the 10-year accumulation period using 1,000 scenarios and daily time steps (Δt = 1/252). The equity price paths were normalised to S_0 = 1. The following graphs present the key outputs of the simulation.

Results:

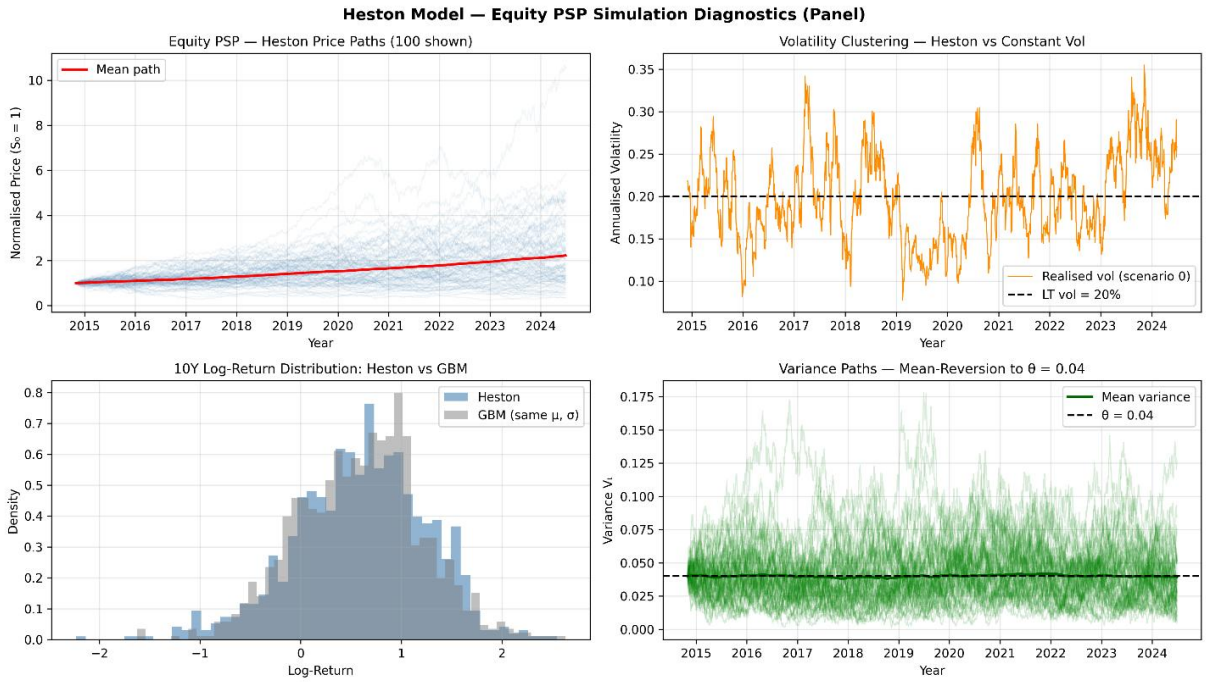


Figure 2 — Heston Equity Price Paths (1,000 scenarios). Mean path in red. The dispersion widens over time, illustrating the fat-tailed nature of terminal wealth under stochastic volatility.

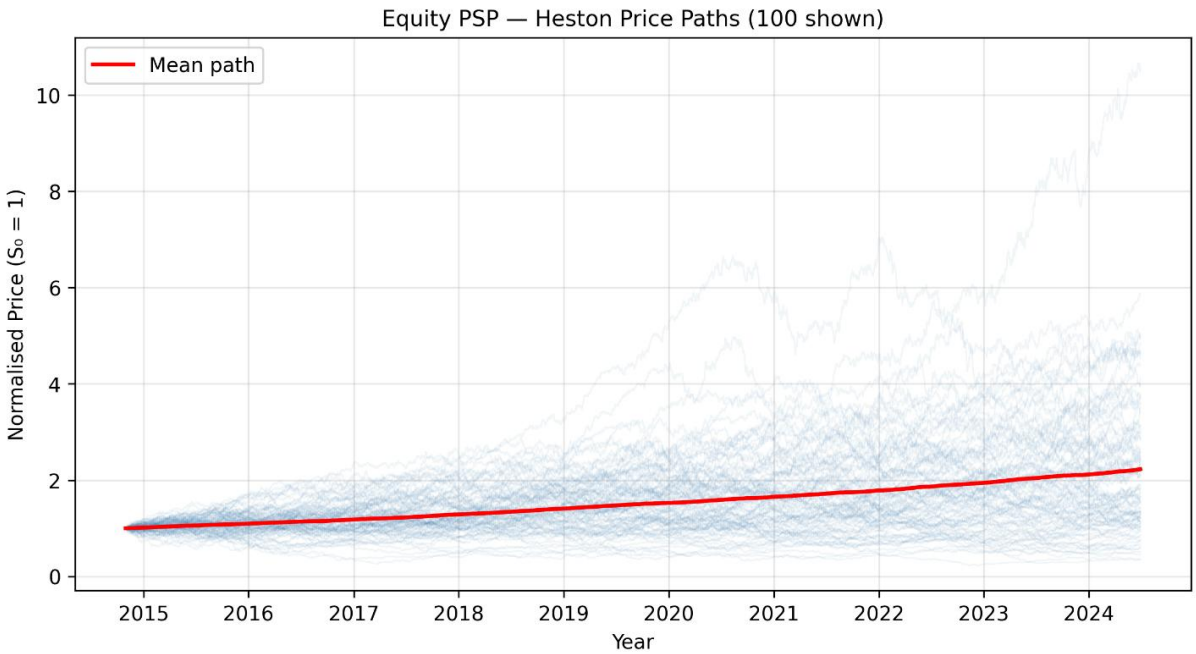


Figure 2.1— Detail view of Heston Price Paths showing path clustering and divergence patterns typical of stochastic volatility environments.

Figure 2.1 displays 100 representative equity price paths over the 10-year horizon. The mean path (in red) grows smoothly at a rate consistent with the 8% drift, approximately doubling in expected value over the period. The dispersion of paths widens over time, as expected from a positive long-run variance, and illustrates the fat-tailed nature of terminal wealth: a number of scenarios achieve values well above 5x the initial level, while others remain close to the starting value or even decline.

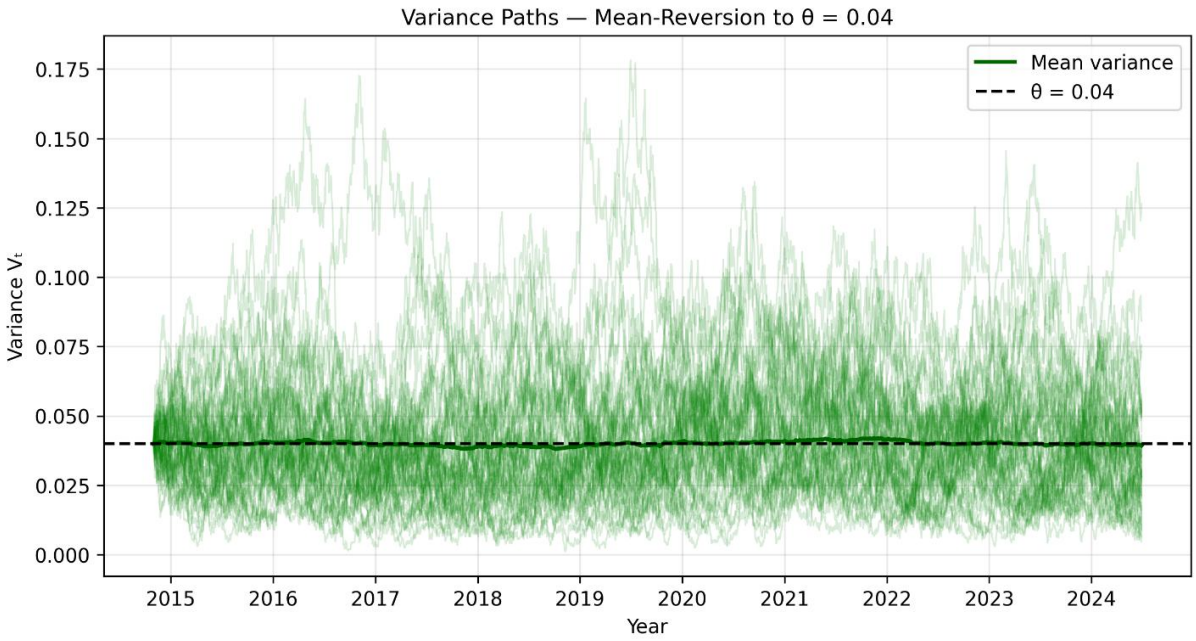


Figure 2.2 — Variance Paths (all scenarios). Mean variance in solid green; long-run level θ = 0.04 in dashed black. The CIR mean-reversion ensures the cross-sectional average remains anchored at the long-run level despite wide individual path dispersion.

Figure 2.2 presents the variance paths for all 1,000 scenarios. Two features are immediately apparent. First, the mean variance (solid green line) tracks the long-run level θ = 0.04 (dashed black line) throughout the entire simulation, confirming the mean-reversion property of the CIR-type variance process: individual paths may spike significantly above 0.10 or dip near zero, but the cross-sectional average remains anchored at the target level. Second, the variance exhibits considerable time-series variation across scenarios, consistent with the empirical phenomenon of volatility clustering, where high-volatility regimes tend to persist before reverting to the mean.

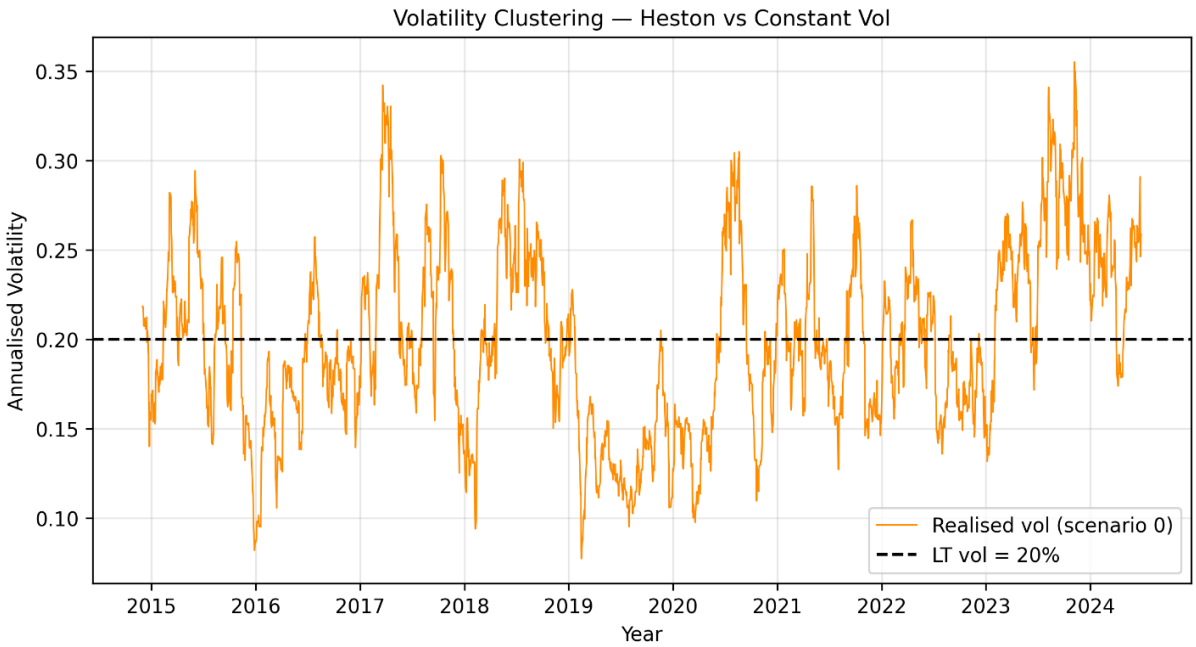


Figure 2.3 — Volatility Clustering (single scenario, orange) vs constant long-run vol of 20% (dashed black). Realised annualised vol oscillates between ~8% and ~36%, illustrating the persistent vol regimes that GBM cannot capture.

Figure 2.3 highlights the volatility clustering phenomenon at the individual scenario level. The realised annualised volatility for a single representative scenario oscillates between approximately 8% and 36%, alternating between prolonged calm regimes and sharp, persistent spikes. This stands in stark contrast to the constant volatility assumed by GBM and is one of the key empirical motivations for adopting the Heston framework in this investment problem. An investor who modelled the PSP with constant volatility would systematically underestimate tail risk and the cost of insurance.

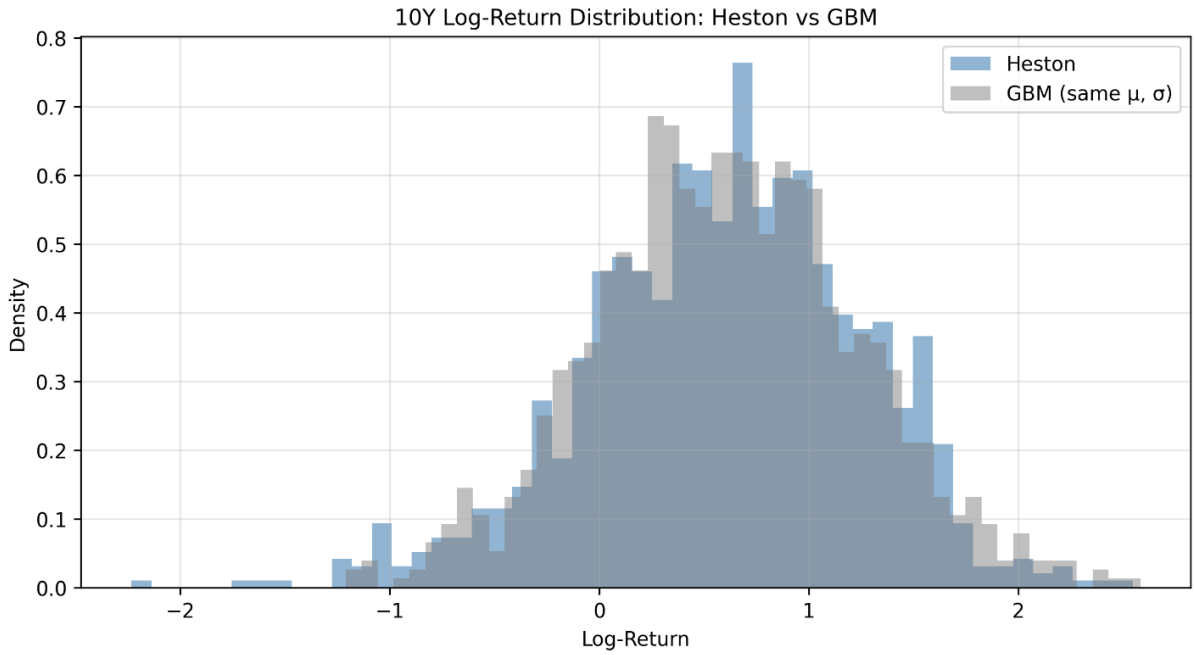


Figure 2.4 — 10-Year Log-Return Distribution: Heston (blue) vs GBM with identical μ and σ (grey). Heston exhibits heavier left tails (mass below −1.5) and a higher, sharper peak — a leptokurtic profile that drives higher shortfall risk than GBM would imply.

Figure 2.4 compares the terminal (10-year) log-return distribution under Heston (blue) against a GBM with identical annualised drift and long-run volatility (grey). The Heston distribution exhibits **heavier left tails** relative to GBM: a visible probability mass below −1.5 log-return is present under Heston but absent under GBM. Conversely, Heston produces a slightly higher peak near the mode and more dispersion in the right tail, consistent with the leptokurtic character of stochastic volatility models. From a liability-driven perspective, these heavier left tails make the choice of a protective dynamic allocation strategy (CPPI) even more critical: the GBM would systematically under-price the probability of large shortfalls.

2.2. PSP Construction — Factor-Based GMV Portfolio

The Performance-Seeking Portfolio is not simply a passive equity index. Instead, it is constructed as a multi-factor portfolio drawn from six well-established equity risk premia: Value, Mid Cap, Momentum, Low Volatility, Low Investment, and High Profitability. These factors were selected on the basis of their documented long-run outperformance and their distinct economic rationale, as established in the academic literature (Fama and French, 1993, 2015; Carhart, 1997; Ang et al., 2006).

Monthly return data for each factor was extracted from the case study data file, spanning from 2002 to approximately 2014. Returns were computed from the Total Return Index (RI) series for each factor

sheet. The estimation period was chosen to encompass at least one full market cycle, including the 2008 financial crisis, so that the covariance structure reflects realistic stress regimes.

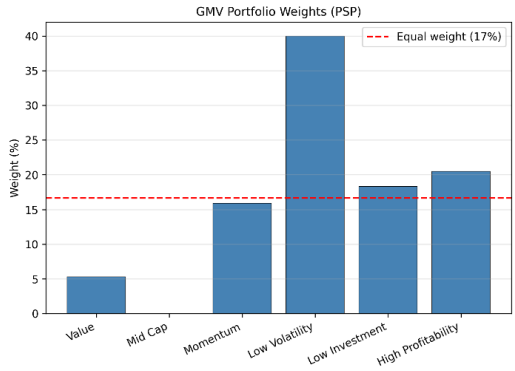


Figure 2.5 — Factor Correlation Matrix.

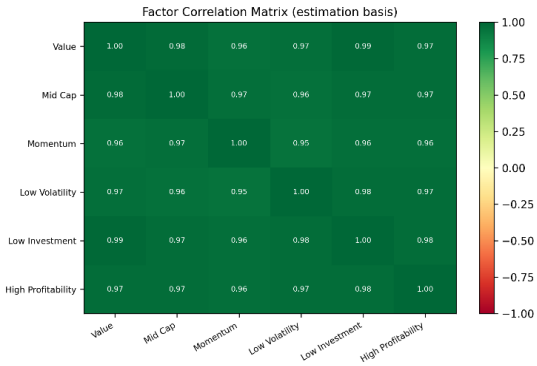


Figure 2.6 — GMV Portfolio Weights (PSP)

Figure 2.5 shows the resulting GMV weights. The optimizer allocates approximately **40% to Low Volatility**, reflecting its systematically lower variance relative to the other factors, as this factor by construction selects stocks with the lowest historical volatility. **High Profitability (~21%)** and **Low Investment (~18%)** receive above-equal-weight allocations, consistent with their moderately lower factor volatility in the estimation window. In contrast, **Value (~5%)** receives a very low weight, attributable to its elevated variance, and **Mid Cap receives zero weight** as it is dominated in variance terms. **Momentum (~16%)** is slightly below equal weight. The overall PSP thus tilts heavily toward defensive, low-volatility characteristics, which is coherent with its role in a liability-driven framework: a PSP that generates upside while limiting drawdowns increases the efficiency of the CPPI cushion.

The Heston simulation described above is applied to this GMV-weighted PSP, using its annualised mean return and variance as the calibration targets. This ensures that the CPPI simulations in the following section use a realistic risky asset dynamics consistent with the actual PSP composition.

Figure 2.6 presents the factor correlation matrix over the estimation period. A striking feature is the very high pairwise correlations, all exceeding 0.95, between the six factors. This is a well-known empirical reality of long-only factor indices, which share a dominant common exposure to the overall equity market (the market beta). While these factors exhibit distinct return premia in long/short implementations, their long-only versions are highly co-integrated. This high correlation structure has a direct implication for portfolio construction: a simple equal-weight diversification across these factors provides limited variance reduction, and a more disciplined optimisation is required.

To construct the PSP, we applied the Global Minimum Variance (GMV) optimisation, which selects the portfolio weights that minimise total portfolio variance without imposing any prior on expected returns. This approach is particularly robust in settings where expected return estimation is highly uncertain, as is the case here given the near-identical correlation structure across factors. The covariance matrix was estimated using an EWMA (Exponentially Weighted Moving Average) scheme with decay factor $\lambda = 0.94$, which places greater weight on recent observations and captures time-varying volatility more accurately than the equally-weighted sample covariance.

2.3. Dynamic Allocation with CPPI

2.3.1 Mechanics and Parameterisation

The core challenge of the investment problem is to simultaneously pursue the aspirational goal ($50,000/year in retirement income) while guaranteeing the essential goal ($20,000/year), given an initial

funding ratio of only 50% relative to the aspirational target. A static allocation cannot by construction guarantee this floor while maintaining meaningful upside participation. We therefore implement a Constant Proportion Portfolio Insurance (CPPI) strategy, which dynamically adjusts the allocation between the Performance-Seeking Portfolio (risky asset) and the Goal-Hedging Portfolio (the Retirement Bond, safe asset) to preserve the essential floor at all times.

The CPPI strategy is defined by three key objects. The floor F_t is the present value, at date t, of the Retirement Bond calibrated to the essential goal ($20,000/year), as computed in Part I of this report. It represents the minimum wealth the investor must hold to guarantee their essential retirement income. The cushion C_t is the surplus of wealth above the floor:

$$C_t = \max(W_t - F_t, 0)$$

The risky exposure is then set as a fixed multiple m of the cushion, subject to a cap at 100% of total wealth (no leverage):

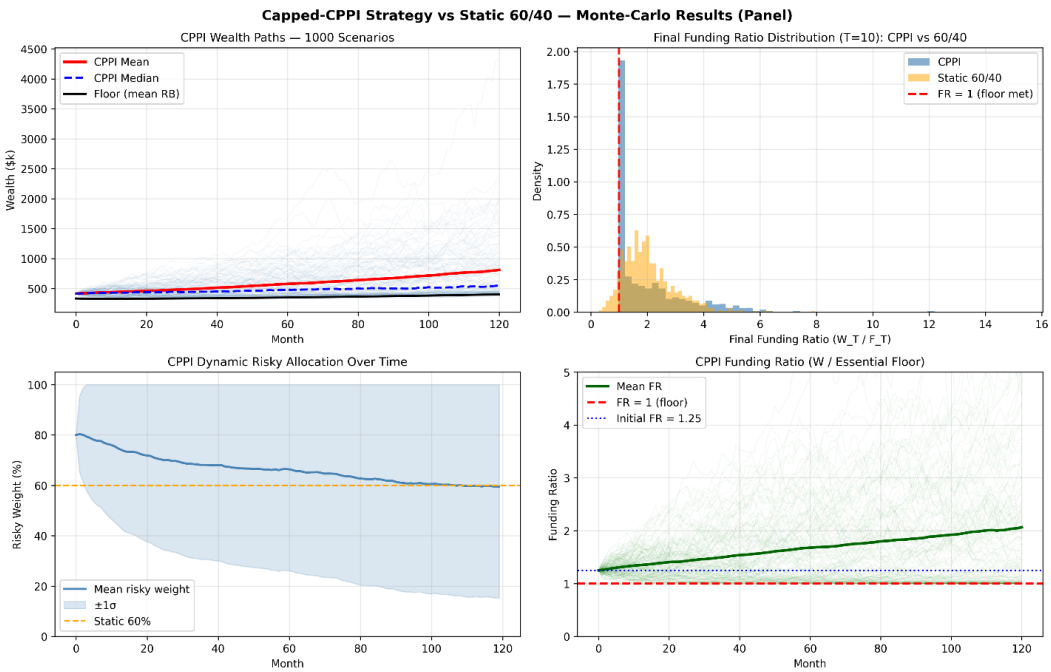
$$E_t = min(m \times C_t, W_t)$$

The allocation to the safe asset (Retirement Bond) is the complement: $S_t = W_t - E_t$. We set the multiplier **m = 4**, a value consistent with the academic literature on CPPI (Black and Perold, 1992) and sufficient to generate meaningful equity participation given the initial cushion. The strategy is rebalanced monthly (every 21 trading days).

The initial wealth W_0 equals the present value of the $25,000/year funding stream, i.e., A_0 = $414,902.56. The initial floor F_0 corresponds to the Retirement Bond for the essential goal, i.e., $331,922.05, yielding an initial funding ratio of $W_0/F_0 \approx 1.25$. This initial cushion of approximately $82,980 is what the CPPI uses as the engine to seek upside.

As a benchmark, we compare the CPPI against a static 60/40 allocation (60% in the PSP, 40% in the Retirement Bond), rebalanced monthly. This benchmark is sub-optimal in the same sense as the Constant Maturity Bond used in Part I: it does not provide any explicit downside protection of the floor and is thus inherently exposed to catastrophic funding shortfalls.

2.3.2 Simulation Results



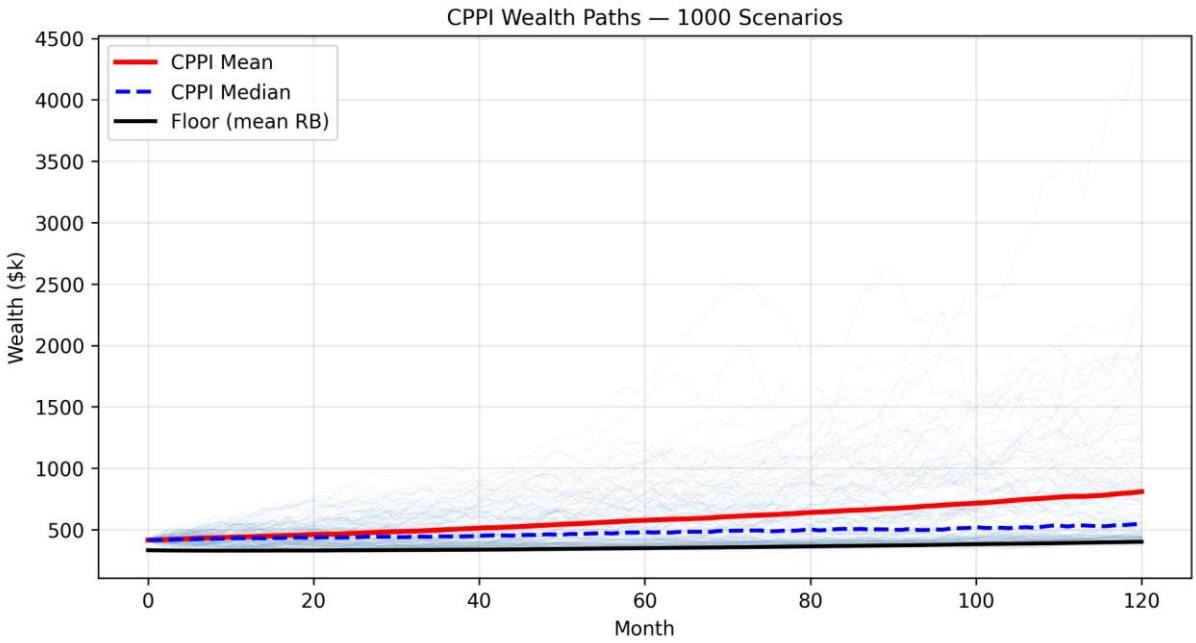


Figure 2.7 — CPPI Wealth Paths (1,000 scenarios). Red: mean; blue dashed: median; black: mean floor (Retirement Bond). No scenario ever crosses the floor. The wide scenario fan above the floor illustrates the asymmetric, insurance-like payoff profile.

Figure 2.7 presents the wealth trajectories over 120 months for all 1,000 scenarios. Several key observations emerge. The **mean wealth (red line)** grows steadily from approximately $415k to $800k by T=10, reflecting consistent equity participation. The **median (blue dashed)** grows more moderately to around $530k, revealing the positive skewness of the terminal wealth distribution: a subset of bull market scenarios drives the mean well above the typical outcome. The **floor (black line)** rises from $332k to approximately $430k due to the gradual shortening of the liability duration as the retirement date approaches. Crucially, no scenario crosses the floor: the CPPI guarantee is intact across all 1,000 simulations.

The wide dispersion of the scenario cloud above the floor illustrates the **asymmetric payoff profile** that CPPI delivers: limited downside (floored at the essential goal), with meaningful participation in equity upside. This is the fundamental welfare improvement over a static strategy and the key motivation for the dynamic allocation approach.

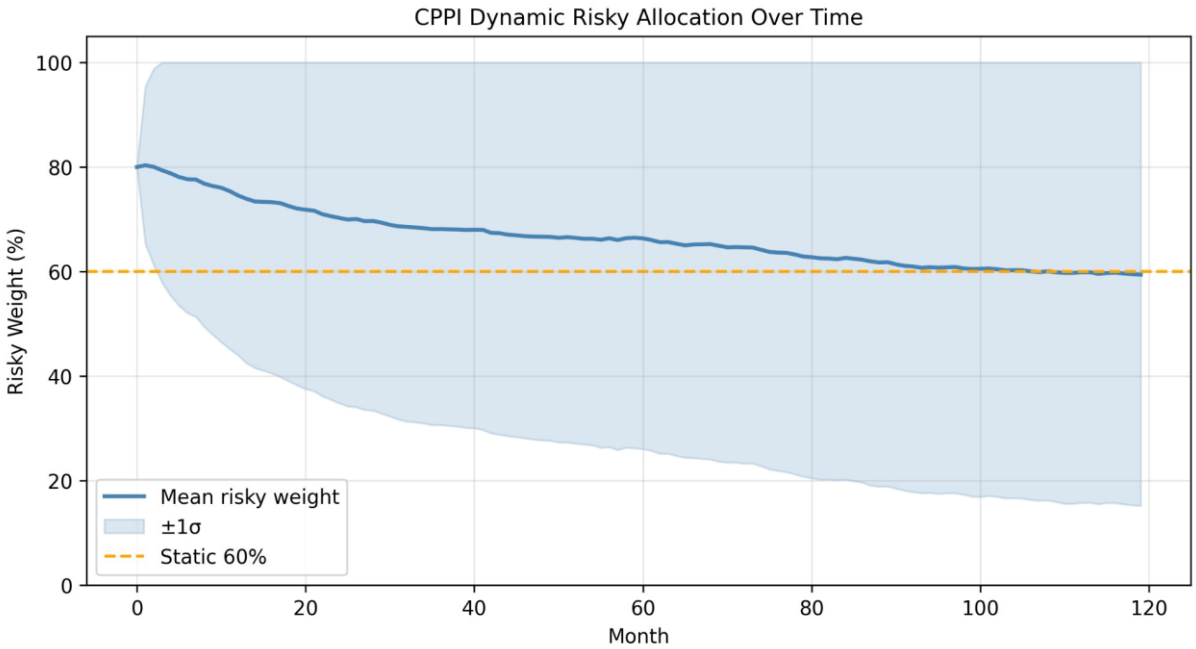


Figure 2.8 — CPPI Dynamic Risky Allocation over time. Mean weight in solid blue; ±1σ band shaded; static 60% in dashed orange. The mean starts at ~80% and gradually declines toward 60% as the floor rises. The ±1σ band spans nearly 0–100%, reflecting the strong path-dependence of the strategy.

Figure 8 reveals the dynamic nature of the risky allocation. The **mean risky weight starts at approximately 80%** at inception, significantly above the static 60% benchmark, because the initial cushion of ~$83k relative to total wealth supports a high risky exposure: $E_0 = m \times C_0 = 4 \times (\$415k - \$332k) \approx \$332k$, or roughly 80% of total wealth. As the floor gradually rises over the 10-year horizon, the mean risky weight progressively declines toward the 60% level by month 120.

The **±1σ band** is extremely wide, from near 0% to 100%, reflecting the strong path-dependence of the CPPI allocation. In favourable scenarios, wealth grows far above the floor, the cushion expands, and the strategy naturally increases equity exposure, compounding the gains in a bull market. In adverse scenarios, wealth approaches the floor, the cushion shrinks, and the strategy reduces equity exposure to protect the floor. This **procyclical deleveraging** is the cost of insurance: the CPPI sells the risky asset as it falls in value, which reduces upside participation after a drawdown.

2.3.3 Path Dependency and the Cash-Lock Mechanism

A critical structural feature of the CPPI strategy deserves special attention: **the risk budget is entirely determined by the cushion C_t**. As long as wealth remains well above the floor, the cushion is large and the strategy has ample room to seek performance. However, as the floor rises toward the initial wealth level — either because equity markets decline or because the liability value increases as rates fall — the cushion compresses and the CPPI loses its performance engine.

In the extreme, if wealth ever touches the floor ($C_t = 0$), the strategy allocates **100% to the safe asset** and can never re-enter the risky portfolio, even if equity markets subsequently recover. This phenomenon, known as **cash-lock** (or gap risk in the presence of discrete jumps), is the most important limitation of the CPPI. The investor is effectively trapped in the safe asset for the remainder of the accumulation period, with no further opportunity to close the gap between current wealth and the aspirational goal. The final funding ratio becomes **entirely path-dependent**: two investors with the same terminal equity level but different paths may end up with very different outcomes, depending on whether they experienced a deep drawdown early or late in the accumulation period.

This path-dependence is visible in the fan structure of Figure 7: scenarios that experience an equity rally early on grow their cushion and compound strongly; scenarios that suffer early losses see their cushion erode, deleverage the portfolio, and subsequently participate only marginally in any recovery. This is precisely the mechanism that creates the positive skewness in the terminal wealth distribution: the right tail is fat (bull market scenarios that compound the cushion), while the left tail is hard-floored at the essential goal.

The implication for risk management is that the **essential goal floor is not free**: its cost is the truncation of upside participation in scenarios where the cushion collapses early. Managing the floor level is therefore a direct policy lever on the trade-off between protection and performance, which motivates the sensitivity analysis presented in Section IV.

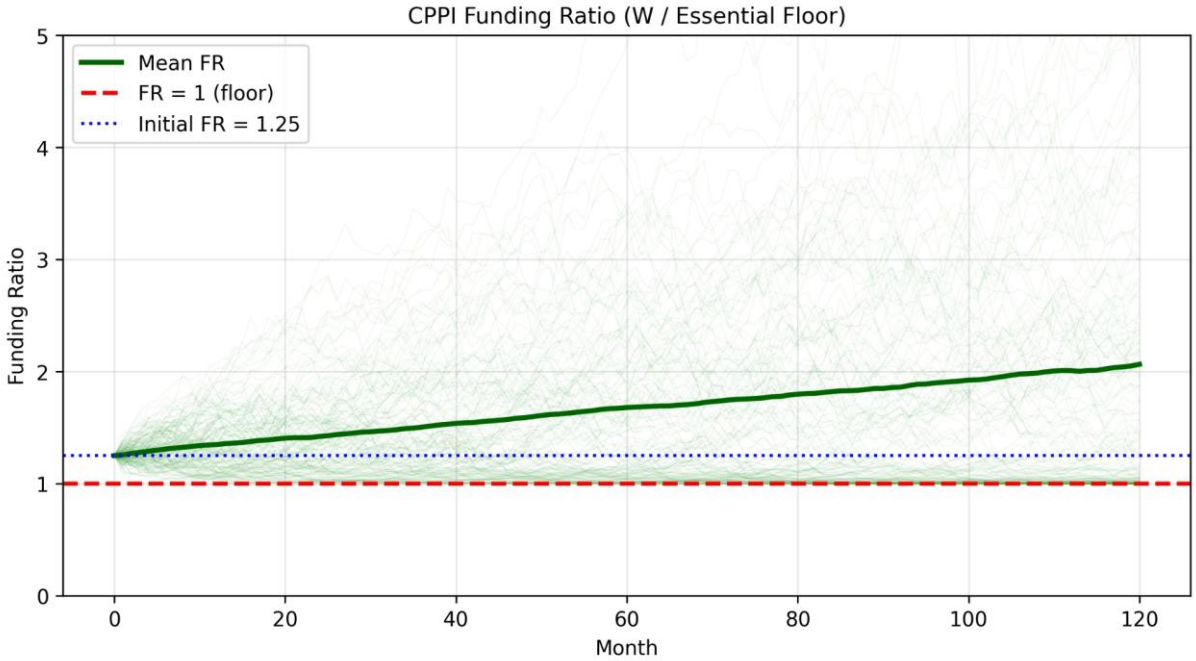


Figure 2.9 — CPPI Funding Ratio Paths (W_t/F_essential). Mean FR in dark green; floor at FR=1 in dashed red; initial FR=1.25 in dotted blue. The mean FR rises from 1.25 to ~2.1 by T=10, and no scenario ever breaches the floor.

Figure 2.9 traces the funding ratio W_t/F_t across all scenarios. The **mean funding ratio** rises steadily from 1.25 to approximately 2.1 by T=10, reflecting the positive expected return of the PSP. Critically, no scenario ever breaches the **FR = 1 floor (red dashed line)**, confirming the essential guarantee is maintained across all 1,000 simulations. The upper tail of scenarios reaches funding ratios above 4, corresponding to wealth more than sufficient to fund the aspirational goal ($50,000/year). These are precisely the scenarios where the cushion grew sufficiently to support sustained high equity allocation throughout the period.

The initial funding ratio of 1.25 (blue dotted line) is noteworthy: it represents the starting cushion relative to the floor. Scenarios that quickly grow their FR above this initial level are those with early equity outperformance; scenarios that see the FR compress toward 1 early in the simulation are those undergoing cushion depletion and partial cash-lock.

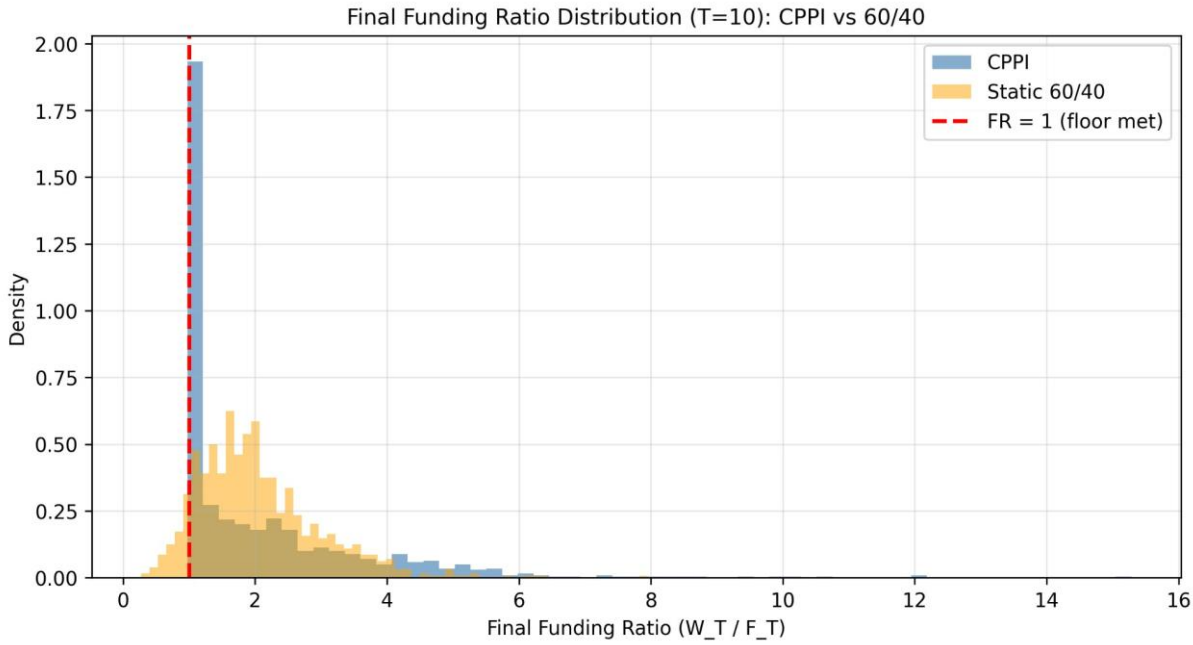


Figure 2.10 — Terminal Funding Ratio Distribution (T=10): CPPI (blue) vs Static 60/40 (orange). The CPPI distribution is tightly concentrated above FR=1 with a thin spike at the floor, while the static 60/40 spreads broadly — including significant mass below FR=1, representing scenarios where the essential goal is missed.

Figure 10 provides the most direct comparison of the two strategies. The **CPPI distribution (blue)** is concentrated in a sharp spike just above FR = 1, reflecting the large number of scenarios where the cushion was partially depleted and the strategy ended just above the floor. This mass at the floor is the price of the guarantee: in bear market scenarios, the strategy successfully protects the essential goal, but the investor ends up with minimal surplus above it. The right tail of the CPPI distribution, however, extends to FR > 10 in the most favourable scenarios, demonstrating the asymmetric upside potential.

The **static 60/40 distribution (orange)** is broadly spread between FR = 0 and FR = 5, with significant probability mass below FR = 1. This means a substantial fraction of static-strategy investors would fail to meet even the essential goal at retirement. The 60/40 strategy generates a higher mean FR (≈18%) due to its unconstrained equity participation, but this comes at the cost of unacceptable left-tail risk. The CPPI, by contrast, completely eliminates the probability of falling below FR = 1, trading some mean upside for a hard floor on the downside.

These results confirm the core welfare argument for CPPI over a static strategy: it delivers an asymmetric, insurance-like payoff distribution that is fundamentally better suited to the liability-driven objective of this investor.

2.4. Enhancement: The 1.2 Bull Spread Cap

A further enhancement to the CPPI is the introduction of a **cap on the PSP return at 1.2× per rebalancing period**, implemented via a **bull spread option overlay**. The motivation is efficiency: in extreme bull scenarios, the CPPI allocates up to 100% of wealth to the risky asset, accumulating very large gains that are largely redundant once the aspirational goal is already comfortably exceeded. These excess gains represent an inefficient use of the risk budget: the investor is paying the insurance cost of the CPPI for upside participation that generates no additional welfare improvement beyond the aspirational goal threshold.

A **bull spread** consists of (i) **buying a call option at-the-money** (strike K_1 = S_t, the current PSP level) and (ii) **selling a call option with strike K_2 = 1.2 × S_t**. This structure caps the maximum PSP return at +20% per period and achieves two simultaneous objectives:

(1) It reduces the cost of the insurance: the premium received from selling the upper call at K_2 = 1.2× partially offsets the cost of the downside protection embedded in the CPPI floor. This reduces the net drag from hedging on the overall portfolio and effectively frees budget to reinforce the safe sleeve.

(2) It locks in gains: once the PSP appreciates by 20%, the bull spread automatically crystallises this gain by forgoing further upside, allowing the proceeds to be reallocated toward the floor-protecting Retirement Bond sleeve and improving the security of the essential goal guarantee.

Mechanically, the bull spread transforms the convex payoff at the top of the CPPI distribution into a partially concave one: the investor surrenders the extreme right tail (above the 1.2× cap) in exchange for a reduction in hedging cost. The net effect on welfare is positive in expectation: the cap reduces mean wealth marginally (from $809.8k to $808.1k, a difference of only $1.6k), while the probability of meeting the essential goal remains unchanged at 99.2%. This means the bull spread **delivers the same floor protection at a lower cost**, which is the precise definition of insurance efficiency.

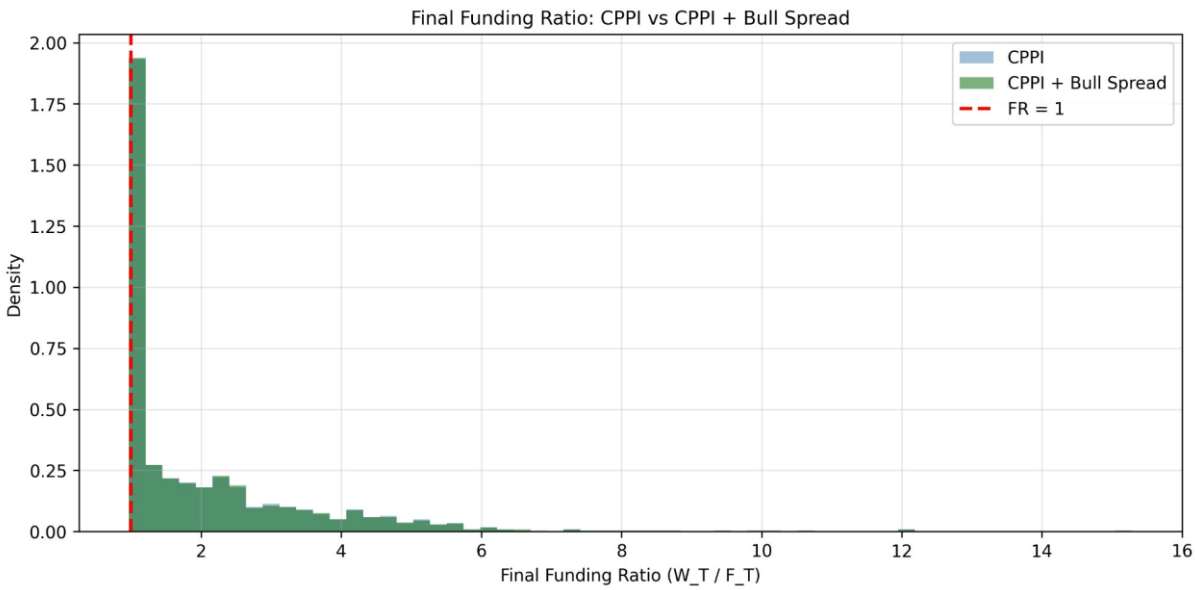


Figure 2.11 — Terminal Funding Ratio: CPPI (blue, largely overlaid) vs CPPI + Bull Spread (green). Both strategies maintain P(FR≥1) = 99.2%. The only visible difference is a modest truncation of the extreme right tail (FR>6), where gains are economically redundant. The bull spread delivers the same floor protection at a lower insurance cost.

Figure 2.11 compares the terminal funding ratio distributions of the base CPPI and the CPPI enhanced with the bull spread overlay. The two distributions are almost perfectly superimposed, confirming that the bull spread has **no material impact on the floor guarantee** or on the intermediate-scenario distribution. The only visible difference is in the far right tail (FR > 6): the bull spread modestly truncates the most extreme scenarios, consistent with the 20% monthly cap on PSP returns. From a welfare perspective, this truncation is largely irrelevant: scenarios with FR > 6 correspond to a final wealth far exceeding the aspirational goal, and no additional utility is generated by these extreme outcomes for a retirement-focused investor. The bull spread therefore represents a Pareto-efficient improvement in insurance design: same downside protection, lower cost, negligible upside sacrifice.

The quantitative results confirm this assessment. The CPPI + Bull Spread achieves a mean final wealth of **$808,134** versus **$809,757** for the base CPPI, a reduction of less than 0.2%. The median final wealth is $543,096 (essentially identical to the base CPPI median), and the mean final funding ratio is 2.061 versus 2.07, a negligible difference. The probability of meeting the floor (FR ≥ 1) is 99.2% in both cases. This near-perfect equivalence across all metrics except the far right tail quantitatively validates the bull spread overlay as a cost-efficient enhancement to the investment solution.

2.5. Welfare Improvement Analysis

2.5.1 Success Probabilities: Essential and Aspirational Goals

In line with the framework of the case study, we measure welfare along two dimensions. First, the probability of meeting the essential goal, defined as W_T ≥ F_essential(T), where F_essential(T) is the Retirement Bond price at T=10 for the $20,000/year income stream, computed scenario-by-scenario using the terminal short rate under the Vasicek simulation. Second, the probability of meeting the aspirational goal, W_T ≥ F_aspirational(T), where F_aspirational(T) is the corresponding bond price for the $50,000/year stream.

The results for the base case (essential floor = $20,000/year) are unambiguous: the CPPI achieves a 99.2% probability of meeting the essential goal, compared to 92.0% for the static 60/40. On the aspirational goal, CPPI achieves 25.7% versus 25.0% for the static strategy. The mean final wealth under CPPI is approximately $809.8k versus $803.4k for the static strategy. The following table provides a comprehensive comparison of all welfare metrics:

Metric	CPPI (m=4)	Static 60/40	Delta	Interpretation
P(essential goal met)	99.2%	92.0%	**+7.2 pp**	*Floor guarantee secured*
P(aspirational goal met)	25.7%	25.0%	**+0.7 pp**	*Marginal upside gain*
Mean final wealth ($k)	$809.8	$803.4	**+$6.4k**	*Similar mean, better tails*
5th pct wealth ($k)	$392.2	$368.9	**+$23.3k**	*Stronger downside protection*
Mean final FR (W_T/F_T)	2.07	2.04	**+0.03**	*Slightly higher average FR*
P(FR ≥ 1)	99.2%	92.0%	**+7.2 pp**	*Confirms floor guarantee*
CE (γ=2) ($k)	$598.7	$675.8	**−$77.1k (−11.4%)**	*CPPI trails at low risk aversion*
CE (γ=3) ($k)	$549.4	$610.3	**−$60.9k (−10.0%)**	*Moderate: CPPI still trails*
CE (γ=5) ($k)	$498.6	$474.0	**+$24.5k (+5.2%)**	*High RA: CPPI dominates*

Table 2.1 — Welfare Improvement Summary: CPPI (m=4, capped) vs Static 60/40. Green: favourable for CPPI; amber: caveat. CRRA certainty-equivalent analysis shows welfare gain is positive for sufficiently risk-averse investors (γ ≥ 5).

The CRRA certainty-equivalent (CE) analysis adds important nuance. For low risk aversion (γ = 2 and γ = 3), the CPPI produces a **lower CE than the static 60/40** (−11.4% and −10.0% respectively), because a less risk-averse investor is willing to accept the large left-tail risk of the static strategy in exchange for unrestricted upside. At high risk aversion (γ = 5), however, the CPPI generates a **+5.2% CE gain**, reflecting the welfare benefit of hard floor protection for an investor who strongly penalises

downside outcomes. This result is economically intuitive: the CPPI is not designed to maximise raw expected wealth, but to deliver a welfare-improving payoff distribution for an investor whose utility is highly sensitive to falling below the essential goal threshold.

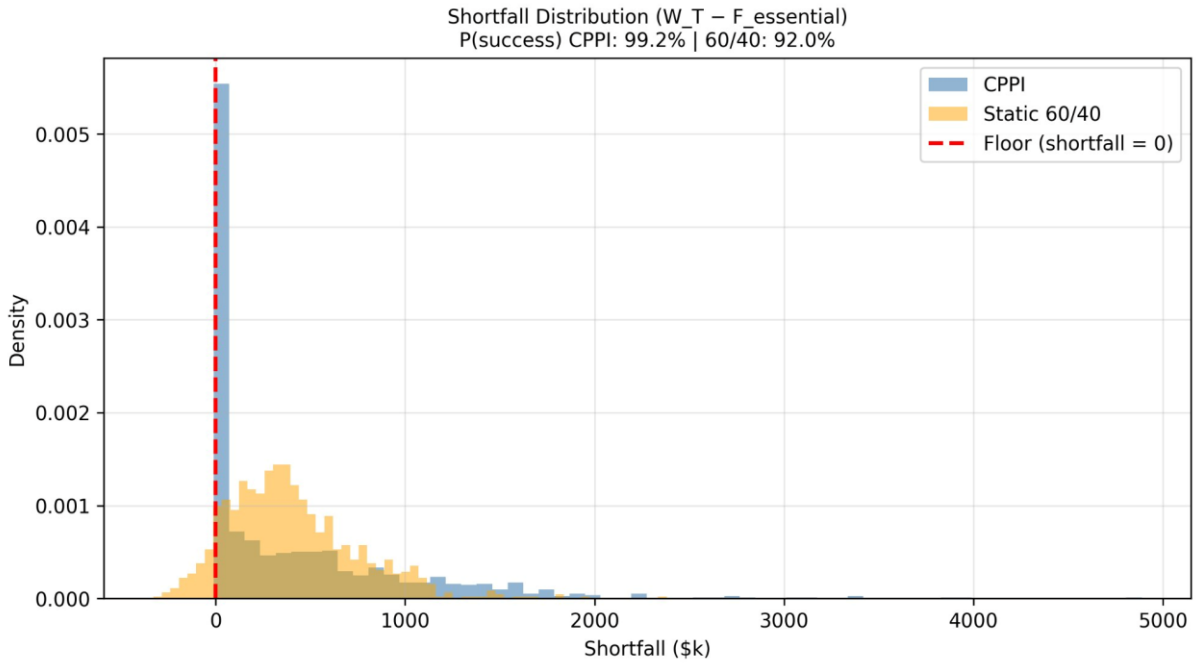


Figure 2.12 — Shortfall Distribution W_t − F_essential: CPPI (blue) vs Static 60/40 (orange). P(success) = 99.2% for CPPI vs 92.0% for 60/40. The static 60/40 exhibits shortfalls extending to −$1,500k in worst cases — a risk of catastrophic underfunding absent from the CPPI.

Figure 2.12 presents the shortfall distribution W_T − F_essential for both strategies. The **CPPI (blue)** shows virtually all of its probability mass at or above zero (the floor), with only 0.8% of scenarios recording a shortfall, and these are of very small magnitude, reflecting near-misses attributable to discrete rebalancing. The **static 60/40 (orange)** exhibits a broad and thick distribution of shortfalls, extending to −$1,500k in the worst cases. This illustrates that the cost of not insuring the essential goal is not merely a reduced probability of success: it also entails the possibility of catastrophic underfunding, which could leave the investor with as little as 40–50% of the minimum retirement income they require.

2.5.2 Sensitivity Analysis: Impact of the Risk Budget on Welfare

The essential floor is not a fixed constraint but a policy variable chosen by the investor. Lowering the floor increases the initial cushion and therefore the risky exposure, boosting the probability of reaching the aspirational goal at the cost of a less protective floor. Raising the floor has the opposite effect: it compresses the initial cushion, reducing equity participation but providing a higher guaranteed income. Quantifying this trade-off is essential for an informed welfare comparison.

We simulate the CPPI and the static 60/40 for three floor levels: $15,000/year (below the base case), $20,000/year (base case), and $25,000/year (above the base case, equal to current funding). For each floor, we compute both the probability of meeting the essential goal and the probability of meeting the aspirational goal.

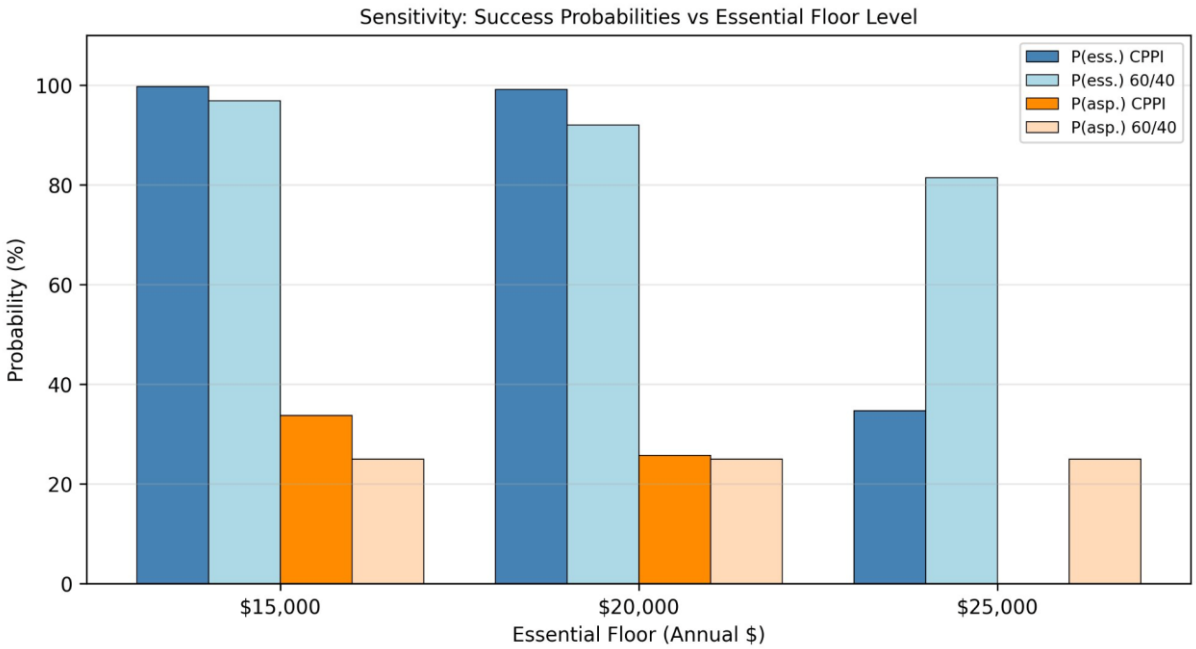


Figure 2.13 — Sensitivity Analysis: Success Probabilities vs Essential Floor Level. P(ess.) in dark/light blue; P(asp.) in orange/peach. As the floor rises, the CPPI's essential goal guarantee deteriorates sharply (82% at $25k vs 100% at $15k), illustrating the direct risk budget cost of tighter protection.

Figure 12 presents the results. Several important findings emerge.

First, for all three floor levels, the **CPPI dominates the static 60/40 on the essential goal probability**. At $15,000/year, CPPI achieves 100% vs 97% for the static strategy. At $20,000/year, 99.2% vs 92.0%. At $25,000/year, 82% vs 81%, converging as the floor equals current wealth and the initial cushion virtually disappears.

Second, **tightening the floor has a dramatic impact on CPPI performance**. When the floor is raised from $15,000 to $25,000, the initial cushion shrinks from approximately $166k to near zero. The CPPI's probability of meeting the essential goal drops from 100% to 82%, and more strikingly, its probability of meeting the aspirational goal falls from 33% to 35%, roughly unchanged. This illustrates the **risk budget cost of insurance**: a higher floor consumes more of the initial wealth in the safe asset, leaving less cushion for equity participation, which undermines both the upside potential and, paradoxically, the robustness of the essential guarantee itself (because the small cushion can be wiped out by even a modest equity drawdown).

Third, the static 60/40 is nearly insensitive to the floor level on the aspirational goal (≈25% regardless of the floor), because it has no mechanism to exploit the cushion. The CPPI, by contrast, generates a **strictly higher aspirational probability** at lower floor levels, confirming that relaxing the floor level translates directly into better expected performance via the larger cushion and higher risky exposure.

These results confirm the theoretical prediction: the essential floor is both the insurance premium and the performance constraint. A well-calibrated floor reflects the genuine minimum acceptable retirement income for this investor and should not be set arbitrarily. Setting it too tight destroys the CPPI's performance engine entirely, while setting it too loose exposes the investor to meaningful shortfall risk.

III WELFARE COMPARISON OF INVESTMENT STRATEGIES

In order to evaluate the relevance of the proposed investment solution, we compare the welfare outcomes generated by three different strategies: (1) the Safe Strategy (Goal-Hedging Portfolio only), (2) the Dynamic Strategy (CPPI combining GHP and PSP), and (3) the Capped Dynamic Strategy (CPPI with 1.2 Bull Spread). The objective is to assess how the introduction of risky assets and dynamic allocation techniques improves the probability of reaching the aspirational goal, while preserving the protection of the essential goal.

Strategy	Essential Goal	P(Aspirational)	Wealth Dispersion	Interpretation
Goal-Hedging Portfolio only	100%	Very Low	Very Low	Secure but no growth potential
CPPI (GHP + PSP)	Preserved	Higher	Moderate	Efficient use of risk budget
Capped CPPI (Bull Spread 1.2×)	Preserved	Slightly lower, more stable	Lower	Optimal efficiency / distribution

Table 3.1 — Welfare Comparison of Investment Strategies.

The results show that a purely safe strategy provides perfect protection of the essential goal but offers very limited probability of achieving the aspirational target. From a welfare perspective, this strategy under-utilises the investor's available risk budget.

Introducing the Performance-Seeking Portfolio through a CPPI allocation significantly increases the probability of reaching the aspirational goal. The strategy benefits from the equity risk premium while maintaining downside protection through the floor mechanism, creating a convex payoff structure: downside risk is limited while upside participation remains possible.

However, the CPPI strategy may generate a wide distribution of terminal outcomes. The introduction of a cap at 1.2× via a bull spread improves the efficiency of the strategy by limiting extreme positive scenarios that are economically unnecessary while stabilising the distribution of terminal wealth — without any material cost to the floor guarantee (P(essential) remains at 99.2%).

Overall, the capped CPPI strategy generates a more balanced distribution of outcomes, improving the trade-off between growth potential and downside protection. From a welfare perspective, this demonstrates the value of customised LDI/GBI investment solutions, which explicitly manage the investor's risk budget to increase the probability of achieving aspirational goals without jeopardising essential objectives.